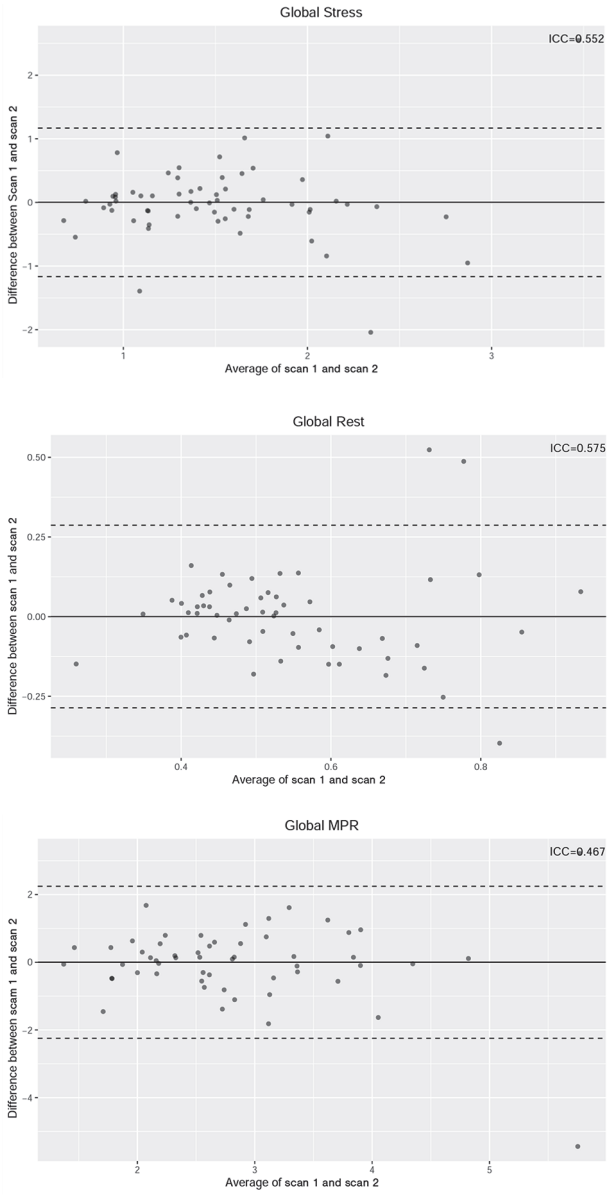

Abstract 16 Figure 1

Conclusion In patients with suspected CAD, quantitative assessment of myocardial perfusion by fully automated inline myocardial mapping shows moderate-good reliability for both stress and rest MBF, but poorer correlation with MPR.

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17 CARDIOVASCULAR MAGNETIC RESONANCE PRESSURE-VOLUME LOOPS ARE FEASIBLE AND HAVE PROGNOSTIC IMPLICATIONS IN HEART FAILURE: A MULTICENTRE UK STUDY

[1,2]Gareth Matthews*, [1,2]Ciaran Grafton-Clarke, [1]Rui Li, [1]Hossam Assadi, [1,2]Zia Mehmood, [2]Faye Nelstrope, [3]Samer Alabed, [2]Alasdair Ryding, [2]Chris Sawh, [3]Andy Swift, [4]Peter Swoboda, [1,2]Pankaj Garg. [1]*Department of Cardiovascular Medicine, University of East Anglia, Bob Champion Research and Education Building, Research Park, Rosalind Franklin Rd, Norwich, Norfolk, UK*; [2]*Department of Cardiology, Norfolk and Norwich University Hospitals Foundation Trust, Colney Lane, Norwich, Norfolk, UK*; [3]*Department of Infection, Immunity and Cardiovascular Disease, The University of Sheffield, Sheffield, UK*; [4]*Leeds Institute of Cardiovascular and Metabolic Medicine, University of Leeds, Leeds, UK*

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Introduction Mortality from heart failure (HF) remains unacceptably high.[1] Left ventricular (LV) ejection fraction (EF) is a flawed, surrogate marker of function.[2] Pressure- Volume (PV) loops permit comprehensive study of myocardial energetics, stiffness and loading conditions. PV-loops can be assessed invasively but this is laborious and carries risk.

Materials and Methods A discovery cohort of PREFER-CMR study patients underwent same-day invasive impedance catheter and cardiac magnetic resonance (CMR) studies. CMR-derived PCWP was calculated as previously validated.[3] Custom software constructed invasive beat-by-beat PV-loops (figure 1A&B), which were used to develop non-invasive approximations from CMR volumetrics and peripheral blood pressure. These were calculated in the outcome cohort of mixed HF patients. Univariable and multivariable Cox regression analysis was applied to assess all-cause and 1-year mortality. ROC-curves found the optimal cut-off values for each metric. These were tested in a validation cohort using Kaplan-Meier analysis.

Results In this multicentre study, a total of 1321 patients were recruited (15 in the discovery and 1306 for the outcomes cohort). There was no significant difference between the stroke work (SW) or static end-diastolic compliance (sEDC) between the CMR and invasive studies. For the outcomes study, patients were randomised to the derivation (n=690) or validation cohort (n=616). The mean follow-up duration was 3.54 years and age was 64.7 years. 48% were female. In the derivation cohort, univariable analysis demonstrated parameters of energetics to show the strongest association with mortality (figure 1C). Multivariable analysis selected sEDC, CMR SW and PE for predicting all-cause mortality, whilst sEDC and CMR SW were selected for 1-year mortality. Kaplan- Meier analysis in the validation cohort found the best performing individual parameter was the sEDC (figure 1D). However the multivariable risk scores showed the best performance in both all cause (figure 1E) and 1-year (χ^2=49.87, P<0.001) mortality. 57.3% of the high-risk patients had preserved EF.

Discussion In this internally validated, multicentre study examining hard mortality endpoints, we show that HF assessment should be multifaceted and include meaningful pathophysiological measures. These are amenable to wide adoption at almost no cost. The era of LVEF guiding HF prognosis and treatment is ending.

Conclusion CMR-derived measures of cardiac energetics and myocardial stiffness predict death in HF more accurately than LVEF.

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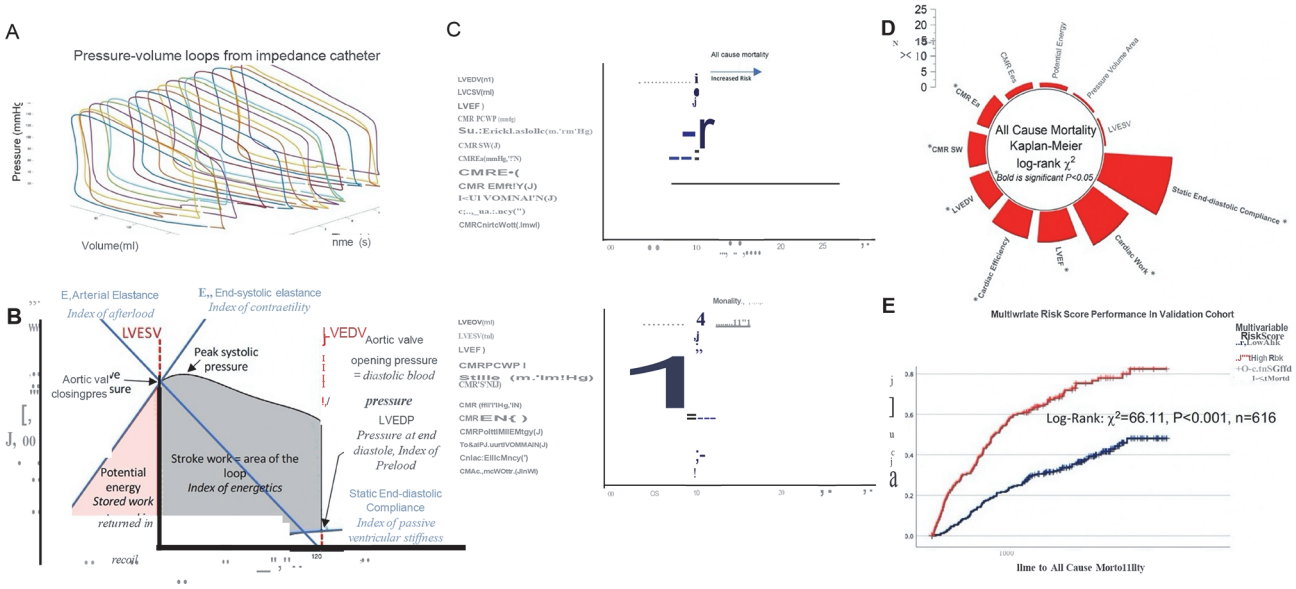

Abstract 17 Figure 1

collating the outcomes data. GM is funded by the National Institute of Health Research. PS is funded by the British Heart Foundation. PG is funded by the Wellcome Trust.

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18 AUTOMATED VENDOR-INDEPENDENT DETECTION AND QUANTIFICATION OF AORTIC STENOSIS FROM A 3-CHAMBER BSSFP CINE VIEW

[1,2,3]Kavitha Vimalesvaran*, [2,3]Sameer Zaman, Antoine Sanchez, [2,3]James P Howard, [4]Henry Proctor, [4]Marilena Giannoudi, [2]Marta Varela, [3]Ben Ariff, [4]Eylem Levelt, [3,5]Nick Linton, [1,5]Anil A Bharath, [2,3]Graham D Cole. [1]*AI for Healthcare Centre for Doctoral Training, Imperial College London, UK SW7 2AZ*; [2]*National Heart and Lung Institute, Imperial College London, UK SW7 2AZ*; [3]*Imperial College Healthcare NHS Trust, London, UK W12 0HS*; [4]*Leeds Institute of Cardiovascular and Metabolic Medicine, University of Leeds, Leeds LS2 9JY, UK*; [5]*Department of Bioengineering, Imperial College London, UK SW7 2AZ*

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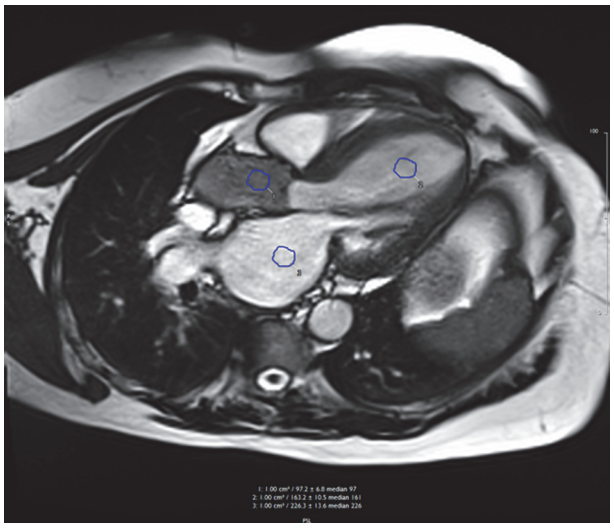
Introduction Cardiac magnetic resonance imaging (CMR) evaluates aortic stenosis (AS) and concurrent myocardial abnormalities. Initial diagnosis of aortic valve (AV) disease can occur during CMR scanning, but universal multiparametric AS assessment is inefficient. An automated strategy for identifying AS patients from the 3-chamber view (part of all protocols) is therefore valuable. We observed that in patients with AS, there was reduced blood signal intensity in the ascending aorta compared to the cardiac chambers. We investigated the diagnostic potential of quantifying this signal reduction as a radiomic marker for AS severity. We introduce an AI-based solution to enhance this metric's precision and reproducibility.

Materials and Methods This was a multi-centre, retrospective cohort study with 249 patients. Signal intensity was quantified within a 1cm^2 region of interest (ROI) in the ascending aorta (Ao), left ventricle (LV), and left atrium (LA) (figure 1). Normalised Ao:LV ratio was calculated and compared against echocardiographic gold standards of AS severity: e.g., dimensionless index (DI) and AV maximum velocity, utilising Pearson correlation coefficients. Automated analysis was conducted using a point-tracking algorithm.

Results The cohort (n = 249, median age 67 [IQR 58–77], 63% male) was stratified into no AS (n=87), mild AS (n=52), moderate AS (n=53) or severe AS (n=57) based on gold-standard echocardiography. The Ao:LV signal ratio strongly correlated with echocardiography parameters (figure 2A-B). A ratio of <0.87 exhibited 87% sensitivity and 83% specificity in identifying aortic stenosis of any severity (figure 2D).



Abstract 18 Figure 1 Example of a 3-chamber cine in a patient with aortic stenosis. A region of interest (ROI) in the aorta, left ventricle and left atrium is manually measured in the end-systolic cardiac phase. The ratio of Ao:LV and Ao:LA bSSFP blood signal in this patient 97/163 = 0.59; 97/226 = 0.43 respectively

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